

README

SRSA MATLAB Code and Reproducibility Examples

This package provides a compact MATLAB implementation of **Super-Resolution Spectral Analysis (SRSA)** and four examples accompanying the manuscript. The examples examine robustness to the assumed model order, resolution of closely spaced frequencies, resolution under Gaussian white noise, and application to a superconducting-gravimeter record.

The numerical workspaces generated by the examples are not distributed because of their large file sizes. To reproduce the numerical results, users should run the corresponding MATLAB scripts. Editable MATLAB figures for the principal results are included as visual references.

Directory organization

```
|-- README.md
|-- src/
|   |-- DoSRSA.m
|-- examples/
|   |-- Demo1_KRobustness.m
|   |-- Demo2_FrequencyResolution.m
|   |-- Demo3_NoiseResolution.m
|   |-- Demo4_M1RealDataApplication.m
|   |-- m1_2011.mat
|-- figures/
    |-- Demo1_KRobustness_Figure (2).fig
    |-- Demo2_FrequencyResolution_Figure (1)_Rules.fig
    |-- Demo2_FrequencyResolution_Figure (1)_Truth.fig
    |-- Demo3_NoiseResolution_Figure (3).fig
    |-- Demo3_NoiseResolution_Figure (4).fig
    |-- Demo4_M1RealDataApplication_Figure (1).fig
```

- `src/` contains the SRSA implementation.
- `examples/` contains the simulation and real-data application scripts. Each script documents the experiment, figure panels, computational environment, and representative runtime in its header comments.
- `examples/m1_2011.mat` is the input data required by Demo 4. It is not a saved output workspace.
- `figures/` contains editable MATLAB figure files for the principal results. These figures permit visual inspection but do not contain the complete numerical workspaces produced by the scripts.

Software requirements

The code was developed and tested with **MATLAB R2024b**. It uses:

- MATLAB;
- Signal Processing Toolbox; and
- Parallel Computing Toolbox (`parfor` in `DoSRSA.m`).

The calculations do not require a GPU. Several examples repeatedly construct or apply Hankel operators and perform singular-value decompositions over multiple values of Q_L ; consequently, a multicore CPU and sufficient RAM are recommended.

The representative runtimes reported in the scripts were obtained using MATLAB R2024b Update 3 on a workstation equipped with an Intel Core i9-14900K processor (24 cores and 32 threads), 128 GB of RAM, and an NVIDIA GeForce RTX 4090 D with 24 GB of memory. A process-based parallel pool with eight workers was used. The current implementation does not invoke GPU computing, so the GPU did not contribute to the reported runtimes.

Quick start

Start MATLAB in the package root directory, or change the MATLAB current folder to this directory, and add the source and example directories to the MATLAB path:

```
rootDir = pwd; % rootDir must be the package root directory
addpath(fullfile(rootDir, 'src'));
addpath(fullfile(rootDir, 'examples'));
```

Run an example by entering its script name, for example:

```
Demo1_KRobustness
```

Demo 4 loads `m1_2011.mat` from `examples/` ; adding the example directory to the path as shown above makes the input file available regardless of the current subdirectory.

The supplied editable figures can be opened without rerunning the calculations. For example:

```
openfig(fullfile(rootDir, 'figures', ...
    'Demo4_M1RealDataApplication_Figure (1).fig'));
```

Opening a `.fig` file displays the supplied figure but does not recreate the complete numerical workspace. Numerical arrays such as the full SRSA responses must be regenerated by running the corresponding example script.

Core function: `DoSRSA.m`

Syntax

```
[Bins,Q_LVEC,IM] = DoSRSA(CObsY,DeltaT,bins,K);  
[Bins,Q_LVEC,IM] = DoSRSA(...,'Q_L',Q_L);  
[Bins,Q_LVEC,IM] = DoSRSA(...,'AlgorithmMode',mode);
```

Inputs

- `CObsY` : real- or complex-valued uniformly sampled time series. Row and column vectors are accepted, and all values must be finite.
- `DeltaT` : positive sampling interval. Its time unit determines the frequency unit of `bins` ; for example, seconds give hertz and years give cycles per year.
- `bins` : candidate frequencies at which the SRSA response is evaluated.
- `K` : assumed signal-subspace dimension, denoted by `TPG` inside the function. It must be a positive integer smaller than the number of samples. For a real-valued sum of sinusoids away from zero and Nyquist frequencies, each sinusoid generally contributes a conjugate pair of complex exponentials; the corresponding ideal signal-subspace dimension is therefore twice the number of tones.
- `'Q_L'` : optional vector of Hankel row counts. Valid values satisfy $1 \leq Q_L \leq N-K$. Invalid requested values are removed with a warning. If omitted, the function selects at most 50 approximately evenly spaced integer values over the valid range.
- `'AlgorithmMode'` : either `'svd'` (default) or `'rsvd'` .

Outputs

- `Bins` : output candidate-frequency vector. For real-valued input it equals `bins` . For complex-valued input, the implementation returns the bilateral frequency grid in reversed-sign order.
- `Q_LVEC` : three-column array `[row_index, Q_L, L]` , where `Q_L` is the number of Hankel rows and `L = N-Q_L+1` is the number of Hankel columns, or window length.
- `IM` : row-wise min-max-normalized SRSA coherence response. Its rows correspond to `Q_LVEC` , and its columns correspond to `Bins` .

Computational modes

- `'svd'` explicitly constructs each Hankel matrix and computes a complete singular-value decomposition. It is deterministic and serves as the reference high-accuracy mode, but has the highest time and memory costs.
- `'rsvd'` uses a matrix-free randomized approximation of the dominant singular subspace. It is more efficient for large problems but may introduce a small approximation error. A fixed geometry-dependent random seed is used internally for reproducibility. If $K \geq \min(Q_L, L)$, the function falls back to the complete-SVD calculation.

Minimal example

```

DeltaT = 1;
t = (0:999)*DeltaT;
y = cos(2*pi*0.10*t) + 0.2*randn(size(t));
bins = 0:0.001:0.2;

[f,QL,response] = DoSRSA(y,DeltaT,bins,2, ...
    'Q_L',round(linspace(100,800,20)), ...
    'AlgorithmMode','svd');

imagesc(f,QL(:,2),response);
axis xy;
xlabel('Frequency (Hz)');
ylabel('Q_L');
colorbar;

```

Interpreting the SRSA outputs

The fundamental SRSA result is the full two-dimensional response $IM(Q_L, f_c)$, rather than a curve at one arbitrarily selected value of Q_L . The parameter Q_L and the corresponding window length $L = N - Q_L + 1$ jointly determine the Hankel-matrix geometry. A useful interval should provide an appropriate compromise between frequency aperture and shifted-sample support.

The displayed products have the following roles:

- **Full 2-D SRS:** the primary result used to inspect the evolution of candidate spectral features over the complete sampled Q_L range.
- **Local 2-D SRS:** a restricted display range used to improve readability within a stable-response interval; it does not replace or override the full 2-D result.
- **Diagnostic cross-section:** a one-dimensional response at a specified Q_L , used to illustrate the spectral shape for that Hankel geometry.
- **Final 1-D SRS:** a compact spectrum obtained by averaging responses over a specified stable Q_L interval.

A candidate peak is regarded as stable when it persists along the Q_L direction, retains a stable frequency position, and can be distinguished from localized background fluctuations or unstable responses near the boundaries of the sampled range.

SRSA peak height is a normalized frequency-coherence response determined by the noise-subspace projection. It is not a physical amplitude, power spectral density, or spectral energy. If physical or relative amplitudes are required, they should be estimated after frequency identification by fitting the identified components to the time-domain data or by applying an appropriate signal-subspace parameter-estimation method.

Examples

Demo 1: robustness to the assumed model order

`Demo1_KRobustness.m` constructs a noisy complex-valued time series containing four complex exponentials at positive and negative frequencies. The script evaluates diagnostic SRSA cross-sections at $Q_L = 1$ and $L = N = 1000$ for

```
K_guess = [2, 7, 10, 50, 100, 200];
```

It also calculates the full 2-D SRSA response using the known model order $K_{\text{true}} = 4$ and displays a normalized bilateral Fourier amplitude spectrum for reference. This example demonstrates bilateral analysis of a complex-valued signal and shows how the principal frequency locations behave when the assumed model order is underestimated or conservatively overestimated.

The noise realization is random by default. Uncomment the supplied `rng(20260025, 'twister')` line if an exactly repeatable realization is required.

Expected products:

- Figure (1): singular-value spectrum;
- Figure (2): six diagnostic cross-sections, the Fourier reference spectrum, and the full 2-D SRSA response.

The package includes the editable Figure (2) as a visual reference. The numerical workspace is generated by running the script and is not distributed.

Representative runtime on the workstation specified above: approximately 1 min.

Demo 2: resolution of closely spaced frequencies

`Demo2_FrequencyResolution.m` analyzes two noise-free, real-valued multitone signals containing five and nine closely spaced sinusoids. Their ground-truth signal-subspace dimensions are $K1_{\text{true}} = 10$ and $K2_{\text{true}} = 18$. Because the true model order is generally unknown in applications, the distributed calculation uses

```
K1_input = 180;  
K2_input = 180;
```

These practical values are selected from the elbow regions of the corresponding singular-value spectra. The ground-truth alternatives $K1_{\text{input}} = 10$ and $K2_{\text{input}} = 18$ are documented in the script. Editable figures for both the rule-based and ground-truth variants are supplied for visual comparison, but their numerical workspaces are not included.

The example displays the simulated series, singular-value spectra, Fourier reference spectra, full 2-D SRSA responses, local 2-D display ranges, and final stacked 1-D SRS results. The local displays use the common interval Q_L in $[8000, 10000]$; the final 1-D

spectra are constructed from the explicitly specified `stackRows`. The complete 2-D responses remain the basis for interpreting the spectral results.

This is a large calculation based on daily sampling over approximately 41 years and dense candidate-frequency grids. Complete-SVD mode can require substantial memory and computation time.

Expected product: Figure (1), containing the simulated signals, singular-value spectra, Fourier spectra, full and local 2-D SRSA responses, and stacked 1-D SRS results.

Representative runtime for the rule-based complete-SVD calculation on the workstation specified above: approximately 25 min.

Demo 3: two-tone resolution in Gaussian white noise

`Demo3_NoiseResolution.m` evaluates two-tone resolution over a nominal 50-year interval. Each time series contains unit-amplitude components at `3 cpy` and `3 + Delta_f cpy`, with

```
Delta_f = [0.040, 0.020, 0.0133]; % cpy
```

The phases are selected so that the two components are in phase at the center of the observation interval. The three frequency separations correspond approximately to the full nominal main-lobe-width reference, one-half of that width, and one-third of that width. Nine groups are formed by combining the three frequency separations with the following pointwise noise-level indices:

```
SNR9999Grid = [1, 5, 9; ...  
               5, 9, 13; ...  
               9, 13, 17];
```

The Gaussian white-noise standard deviation is calibrated from the one-sided Fourier-amplitude normalization. At a fixed nonzero frequency bin, the pointwise 99.99th-percentile Rayleigh noise amplitude is prescribed as `1/SNR9999` of the theoretical amplitude of a single tone.

For each of the nine groups, the script generates 100 independent Monte Carlo realizations, and computes singular-value spectra, Fourier amplitude spectra, and SRSA responses for all 900 realizations. The SRSA calculations use matrix-free randomized SVD. Ensemble averaging reduces realization-specific noise fluctuations and allows assessment of whether the two candidate frequencies remain consistently distinguishable.

Expected products:

- Figure (1): singular-value spectra for the 100 realizations in each group;
- Figure (2): Fourier amplitude spectra for all realizations, together with the prescribed noise-amplitude threshold and theoretical single-tone amplitude;

- Figure (3): full ensemble-mean 2-D SRSA responses for all nine groups;
- Figure (4): selected 2-D SRSA displays and comparisons between the stacked 1-D SRS and ensemble-mean Fourier spectra.

This is the most computationally intensive example. It performs 900 complete singular-value analyses and 900 rSVD-based SRSA evaluations. The resulting workspace can occupy several gigabytes if saved. The package therefore provides editable Figures (3) and (4), but not the generated numerical workspace.

Representative runtime for the complete experiment on the workstation specified above: approximately 70 min.

Demo 4: application to the 2011 M1 superconducting-gravimeter record

`Demo4_M1RealDataApplication.m` demonstrates the data-driven SRSA workflow in the target frequency band of the Earth's fundamental spheroidal mode $0S_2$. The required input file `examples/m1_2011.mat` contains two versions of the M1 record in `series_m1`: the original preprocessed and model-corrected record sampled at 60 s, and a band-pass-filtered and downsampled series sampled at 300 s.

The script first compares the Fourier power spectra of the two series to confirm that the principal spectral structure within the target band is retained after preprocessing. It then examines the singular-value spectrum of the 300-s series and evaluates SRSA for three adjacent model orders selected near its elbow:

```
K_used = [50, 55, 60];
```

For each model order, the script displays the full frequency- Q_L response, the stable interval Q_L in $[641, 2321]$, and the corresponding mean-stacked 1-D SRS. Published normal-mode frequencies are plotted only as interpretive references; they are not used as fixed-frequency constraints, peak-picking criteria, or inputs to the SRSA calculation.

Expected product: Figure (1), containing the time-series and power-spectrum comparisons, singular-value analysis, full and selected 2-D SRSA responses, and mean-stacked 1-D spectra. An editable version of the figure is supplied; the generated numerical workspace is not included.

Representative runtime on the workstation specified above: approximately 1 min.

Reproducibility and computational notes

1. Run the scripts without modification to reproduce the stated experimental designs and numerical workflows. To reproduce the exact Demo 1 noise realization, uncomment the fixed random-seed line provided in that script. Changing the frequency-grid density,

sampled `Q_L` values, model order, Monte Carlo count, random seed, or decomposition mode defines a different numerical experiment.

2. When `'Q_L'` is not supplied, `DoSRSA.m` evaluates at most 50 approximately evenly spaced valid Hankel geometries. Supplying `'Q_L'` provides explicit control and can reduce the cost of exploratory calculations.
3. Complete-SVD results are deterministic. The randomized-SVD implementation uses fixed internal seeds so that repeated calls with identical inputs and settings are reproducible within normal floating-point tolerances.
4. The scripts use `parfor` to distribute calculations over `Q_L`. Runtime depends on the number of workers, processor performance, memory bandwidth, candidate-frequency-grid size, and the number and dimensions of the Hankel operators.
5. Explicit Hankel matrices and complete singular-value decompositions can require substantial memory. For exploratory calculations, the number of candidate frequencies or sampled `Q_L` values may be reduced, or `'rsvd'` may be used when its approximation is acceptable. Any such change should be reported because it modifies the numerical experiment.
6. The scripts create numerical variables and MATLAB figures in the current session. They do not automatically save the generated workspaces or overwrite the supplied `.fig` files.
7. If a generated workspace is saved manually, use MATLAB `-v7.3` format when the variables exceed the standard MAT-file size limit. Demo 3 in particular may produce a multi-gigabyte workspace.
8. MATLAB `.fig` files can be opened with `openfig`. Use MATLAB export functions when a standard raster or vector format is required.

Citation and contact

If this code contributes to published work, please cite the associated SRSA manuscript and identify the code version used.

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